Version:1

Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

USE OF FENTANYL PCA FOR LABOUR ANALGESIA

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PROPRIETARY INFORMATION

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1. Introduction and Purpose

Fentanyl PCA is an useful option for labour analgesia when epidural analgesia is contraindicated, and the patient has been commenced on, but deemed unsuitable to continue on a remifentanil PCA. However, it does not provide as effective analgesia as a fully functioning epidural. For safe use, both the parturient and neonate require careful monitoring both during labour and after delivery.

2. Scope

This guideline is intended as reference for trainee anaesthetists, and midwifes who wish to provide patient controlled fentanyl analgesia for parturients in labour.

3. Points to consider

3.1 Indication

Patient should be in the first stage of labour and not coping with the contraction pain despite analgesia with Entonox and intermittent intramuscular opioid injections. She must also be unable to have an epidural due to it being contraindicated (including maternal refusal). She must also have been commenced first on a remifentanil PCA, which has needed to be discontinued due to apnoeic episodes. It can also be used for women in labour with an intrauterine death.

3.2 Setting up the PCA

- The anaesthetist on call for delivery suite is responsible for prescribing the fentanyl PCA while the midwife is responsible for setting up and maintaining it.
- Use a dedicated Arcomed PCA device
- Draw up 500mcg fentanyl (10mls of 50mcg/ml) and dilute with 40 ml of saline in a 50 ml syringe to make up a 10 mcg/ml solution
- Label the syringe clearly with the drug name, dilutant and concentration
- Connect the extension tubing onto the syringe, and prime the line
- Load the syringe into the PCA device
- Program the PCA device as per standard fentanyl PCA prescription (see below)
- Prescribe the PCA on the patient's prescription chart. The prescription should read :

Fentanyl PCA (500mcg in 50 mls)
Bolus dose : 10 mcg (1ml)
Lockout period : 5 minutes

- Prescribe naloxone (100mcg intravenously every 2 minutes until respiratory rate > 8 /min and consciousness has returned or 400mcg intramuscularly if respiratory rate < 8 breaths /min)

- Connect the PCA to the patient using a Y- connector with a one way valve on the intravenous infusion port

3.3 Monitoring

- While the Fentanyl PCA is being used, the midwife should monitor the patients vital signs, particularly the respiratory rate and level of sedation, at regular intervals during labour

- If the midwife has any concern regarding the function of the Fentanyl PCA or it's effects on the patient, then she should contact the duty obstetric anaesthetist

3.4 Troubleshooting

- Beware of respiratory depression in the newborn baby due to the fentanyl, and have naloxone available. Always notify the paediatricians that the mother has received a Fentanyl PCA during labour

- If the patient's respiratory rate drops below 7 breath/min and/or they become unrousable then :

  i. SUMMON HELP, including calling the anaesthetist
  ii. Remember **A**irway **B**reathing **C**irculation
  iii. Give 6 litres oxygen via facemask if patient is still breathing. Alternatively, if the patient is apnoeic then ensure a patent airway then bag the patient with 100% oxygen using an Ambu Bag)
  iv. Disconnect the PCA device
  v. Give naloxone intravenously or intramuscularly

4. References

1. Fortescue C, Wee MYK. Analgesia in labour : non-regional techniques. *Br J Anaesthesia: CEACCP* (2005) Vol. 5 (1): 9-13